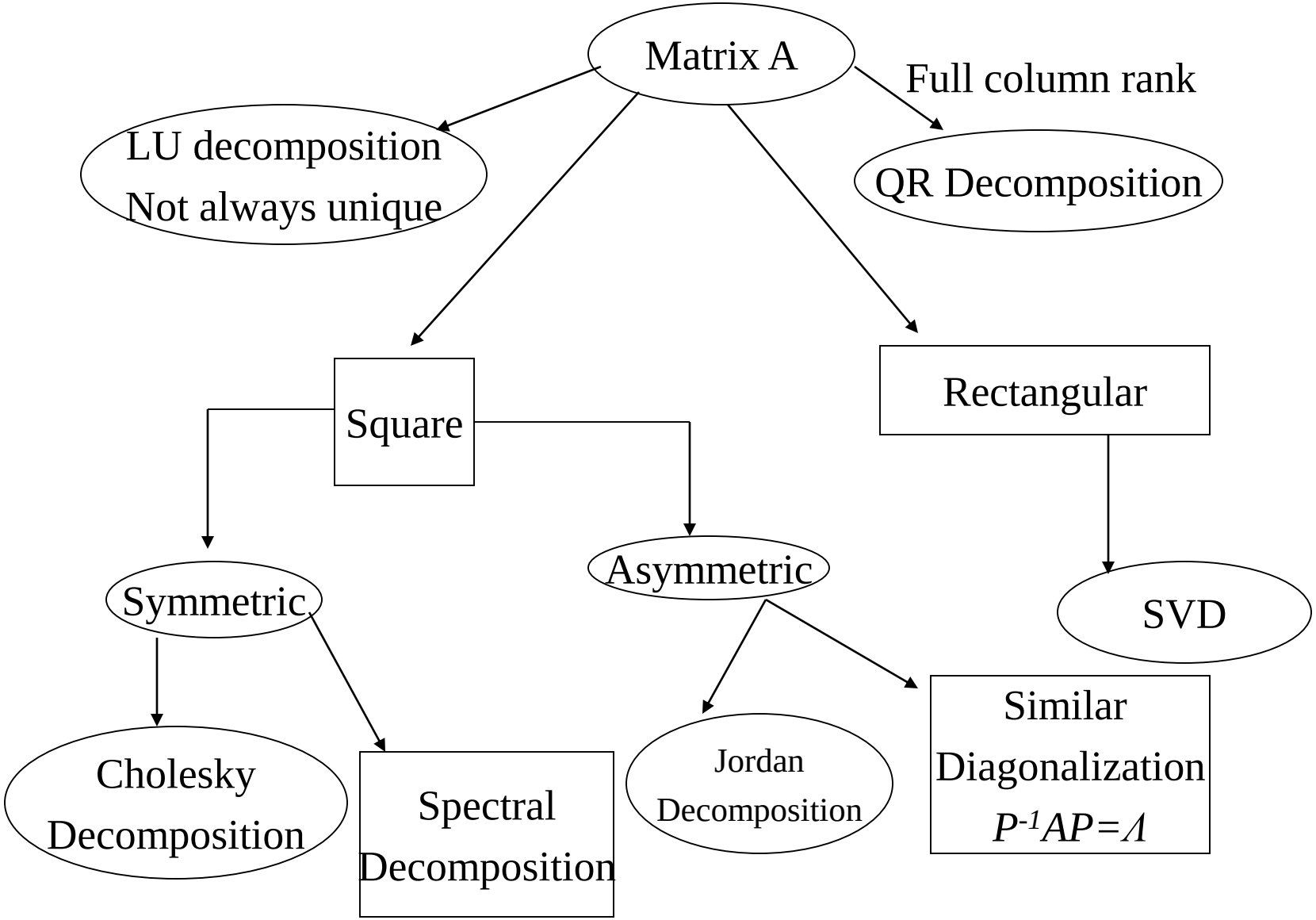
Matrix Decomposition

- Decompositions provide a numerically stable way to solve a system of linear equations, and
- To invert a matrix.
- Additionally, they provide an important tool for analyzing the numerical stability of a system.

Decomposition in Diagram



Easy to solve system

Upper Triangular Matrix: Used in Gauss Elimination Method

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ 0 & a_{22} & a_{23} & \dots & a_{2n} \\ 0 & 0 & a_{33} & \dots & a_{3n} \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$$

Solution: This system is solved using Backward substitution

$$x_i = (b_i - \sum_{j=i+1}^{n} a_{ij}.x_j)/a_{ii}$$

Easy to solve system

Lower triangular matrix: Can be used also

$$\begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ a_{31} & a_{32} & a_{33} & \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$$

Solution: This system is solved using forward substitution

$$x_i = (b_i - \sum_{j=1}^{i-1} a_{ij} . x_j) / a_{ii}$$

Easy to solve system

Some linear system that can be easily solved

$$\begin{bmatrix} a_{11} & 0 & 0 & \cdots & 0 \\ 0 & a_{22} & 0 & \cdots & 0 \\ 0 & 0 & a_{33} & \cdots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \cdots & a_{nn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \\ \vdots \\ b_n \end{bmatrix}$$

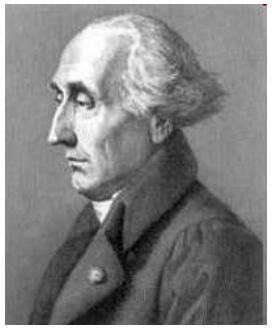
The solution: $\begin{bmatrix} b_1 / a_{11} \\ b_2 / a_{22} \\ \vdots \\ b_n / a_{nn} \end{bmatrix}$

LU Decomposition

LU decomposition was originally derived as a decomposition of quadratic and bilinear forms. Lagrange, in the very first paper in his collected works (1759) derives the algorithm we call Gaussian elimination. Later Turing introduced the *LU* decomposition of a matrix in 1948 that is used to solve the system of linear equation.

Let A be a $m \times m$ with nonsingular square matrix. Then there exists two matrices L and U such that, where L is a lower triangular matrix and U is an upper triangular matrix.

$$U = \begin{bmatrix} u_{11} & u_{12} & \cdots & u_{1m} \\ 0 & u_{22} & \cdots & u_{2m} \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & u_{mm} \end{bmatrix} \quad \text{and} \quad L = \begin{bmatrix} l_{11} & 0 & \cdots & 0 \\ l_{21} & l_{22} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ l_{m1} & l_{m2} & \cdots & l_{mm} \end{bmatrix}$$



J-L *Lagrange*

(1736 –1813)

A. M. Turing

(1912-1954)

Where, $A = LU$

Calculation of L and U *(cont.)*

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix}$$

Now reducing the first column we have

$$\begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & -12 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix}$$

$$\Rightarrow \begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix}$$

Calculation of L and U *(cont.)*

Now $\begin{bmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ -1/2 & 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -3 & 1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 3 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 3 & 1 \end{bmatrix}$

Therefore,

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 3 & 1 \end{bmatrix} \begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix} = \boldsymbol{LU}$$

If A is a Non singular matrix then for each L (lower triangular matrix) the upper triangular matrix is unique but an LU decomposition is not unique. There can be more than one such LU decomposition for a matrix. Such as

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix} = \begin{bmatrix} 6 & 0 & 0 \\ 12 & 1 & 0 \\ 3 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 & -\frac{2}{6} & \frac{2}{6} \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix} = \boldsymbol{LU}$$

Calculation of *L* and *U (cont.)*

- Thus *LU* decomposition is not unique. Since we compute *LU* decomposition by elementary transformation so if we change L then U will be changed, such that A=LU
- To find out the unique *LU* decomposition, it is necessary to put some restriction on *L* and *U* matrices. For example, we can require the lower triangular matrix *L* to be a unit one (i.e. set all the entries of its main diagonal to ones).
- Uses: The LU decomposition is most commonly used in the solution of systems of simultaneous linear equations. We can also find determinant easily by using LU decomposition (Product of the diagonal element of upper and lower triangular matrix).

Solving system of linear equation using LU decomposition

- Suppose we would like to solve a $n \times n$ system $AX = b$. Then we can find a LU-decomposition for A, then to solve $AX = b$, it is enough to solve the systems

$$LY = b$$

$$UX = Y$$

- Thus the system $LY = b$ can be solved by the method of forward substitution and the system $UX = Y$ can be solved by the method of backward substitution. To illustrate, we give some examples
- Consider the given system $AX = b$, where

$$A = \begin{bmatrix} 6 & -2 & 2 \\ 12 & -8 & 6 \\ 3 & -13 & 2 \end{bmatrix} \quad \text{and} \quad b = \begin{pmatrix} 8 \\ 14 \\ -17 \end{pmatrix}$$

Solving system of linear equation using LU decomposition

We have seen $A = LU$, where

$$L = \begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 3 & 1 \end{bmatrix} \quad U = \begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix}$$

Thus, to solve $A\boldsymbol{X} = \boldsymbol{b}$, we first solve $L\boldsymbol{Y} = \boldsymbol{b}$ by forward substitution

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 1 & 0 \\ 1/2 & 3 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 8 \\ 14 \\ -17 \end{bmatrix}$$

Then

$$Y = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ -15 \end{bmatrix}$$

Solving system of linear equation using LU decomposition

Now, we solve $U\boldsymbol{X} = \boldsymbol{Y}$ by backward substitution

$$\begin{bmatrix} 6 & -2 & 2 \\ 0 & -4 & 2 \\ 0 & 0 & -5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ -15 \end{bmatrix} \quad \text{then} \quad \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$$

Method: [A] Decomposes to [L] and [U]

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix} \begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}$$

- [*U*] is the same as the coefficient matrix at the end of the forward elimination step.
- [*L*] is obtained using the *multipliers* that were used in the forward elimination process

Finding the $[U]$ matrix

Using the Forward Elimination Procedure of Gauss Elimination

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

Step 1: $\dfrac{64}{25} = 2.56; \quad Row2 - Row1(2.56) = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 144 & 12 & 1 \end{bmatrix}$

$$\frac{144}{25} = 5.76; \quad Row3 - Row1(5.76) = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{bmatrix}$$

Finding the [U] matrix

Matrix after Step 1: $\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{bmatrix}$

Step 2: $\dfrac{-16.8}{-4.8} = 3.5; \quad Row3 - Row2(3.5) = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$

$$[U] = \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

Finding the $[L]$ matrix

$$\begin{bmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{bmatrix}$$

Using the multipliers used during the Forward Elimination Procedure

From the first step of forward elimination

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix}$$

$$\ell_{21} = \frac{a_{21}}{a_{11}} = \frac{64}{25} = 2.56$$

$$\ell_{31} = \frac{a_{31}}{a_{11}} = \frac{144}{25} = 5.76$$

Finding the $[L]$ matrix

From the second step of forward elimination

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & -16.8 & -4.76 \end{bmatrix}$$

$$\ell_{32} = \frac{a_{32}}{a_{22}} = \frac{-16.8}{-4.8} = 3.5$$

$$[L] = \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix}$$

Does [L][U] = [A]?

$$[L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} = \ ?$$

Using LU Decomposition to solve SLEs

Solve the following set of linear equations using LU Decomposition

$$\begin{bmatrix} 25 & 5 & 1 \\ 64 & 8 & 1 \\ 144 & 12 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Using the procedure for finding the [L] and [U] matrices

$$[A] = [L][U] = \begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix}$$

Example

Set $[L][Y] = [b]$

$$\begin{bmatrix} 1 & 0 & 0 \\ 2.56 & 1 & 0 \\ 5.76 & 3.5 & 1 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ 177.2 \\ 279.2 \end{bmatrix}$$

Solve for $[Y]$

$$y_1 = 106.8$$
$$2.56y_1 + y_2 = 177.2$$
$$5.76y_1 + 3.5y_2 + y_3 = 279.2$$

Example

Complete the forward substitution to solve for $[Y]$

$$y_1 = 106.8$$
$$y_2 = 177.2 - 2.56y_1$$
$$= 177.2 - 2.56(106.8)$$
$$= -96.2$$
$$y_3 = 279.2 - 5.76y_1 - 3.5y_2$$
$$= 279.2 - 5.76(106.8) - 3.5(-96.21)$$
$$= 0.735$$

$$[Y] = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ -96.21 \\ 0.735 \end{bmatrix}$$

Example

Set $[U][X] = [Y]$

$$\begin{bmatrix} 25 & 5 & 1 \\ 0 & -4.8 & -1.56 \\ 0 & 0 & 0.7 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 106.8 \\ -96.21 \\ 0.735 \end{bmatrix}$$

Solve for $[X]$

The 3 equations become

$$25x_1 + 5x_2 + x_3 = 106.8$$
$$-4.8x_2 - 1.56x_3 = -96.21$$
$$0.7x_3 = 0.735$$

Example

From the 3rd equation

$$0.7x_3 = 0.735$$
$$x_3 = \frac{0.735}{0.7}$$
$$x_3 = 1.050$$

Substituting in a_3 and using the second equation

$$-4.8x_2 - 1.56x_3 = -96.21$$
$$x_2 = \frac{-96.21 + 1.56a_3}{-4.8}$$
$$x_2 = \frac{-96.21 + 1.56(1.050)}{-4.8}$$
$$x_2 = 19.70$$

Example

Substituting in x_3 and x_2 using the first equation

$$25x_1 + 5x_2 + x_3 = 106.8$$

$$x_1 = \frac{106.8 - 5x_2 - x_3}{25}$$

$$= \frac{106.8 - 5(19.70) - 1.050}{25}$$

$$= 0.2900$$

Hence the Solution Vector is:

$$\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0.2900 \\ 19.70 \\ 1.050 \end{bmatrix}$$

Assignment:

Given $x_1 = -2$, $x_2 = -1$, $x_3 = 1$ and $x_4 = 2$

1. A 4x4 |A| and a 4x1 |b| matrix like below (or you may make)
2. Make L and U matrix from A using LU decomposition method
3. Show that the values x like above.

$$\begin{vmatrix} 12 & -3 & 6 & 4 \\ 2 & 11 & 3 & 5 \\ 4 & 1 & 10 & 2 \\ -5 & 3 & 4 & -10 \end{vmatrix} \cdot \begin{vmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \end{vmatrix} = \begin{vmatrix} -7 \\ -2 \\ 5 \\ -9 \end{vmatrix}$$